

SHOW YOUR DETAILED SOLUTIONS AND CALCULATIONS - Final answers

with no solution will get ZERO

Carry your intermediate computations to at least three decimal places.

**Q1- Use the golden-section search to find the minimum of $y = x^2 - 4\sin(5x)$
assume that $x_l = 1.3$ and $x_u = 1.7$ (10 marks)**

Answer:

.....

Q2- Solve the integral $\int_0^1 \frac{3}{1+x^2} dx$ by Trapezoidal rule with $n = 1, 2, 4$ and 8. Then obtain the better estimate using Romberg integration method. (13 marks)

Answer:

.....

**Q3- Consider the ordinary differential equation: $2 \frac{dy}{dx} - 3e^x + 2y = 0$
Given: $y(0) = 0, h = 0.5$. Use the Runge-Kutta method to the 4th order to find $y(1)$.
(12 marks)**

Answer:

.....

Q4: From the following data shown in the table below (15 marks)

$x:$	-1	0	1	3	5
$f(x)$	3.1	0	4.3	7.5	2.2

1- Find $f(2)$ - employing Newton's forward interpolation formula.

Answer:

.....

2- Find the area bounded by the curve and x axis from $x = -1$ to $x = 5$ using

Simpson's 1/3 rule

Answer:

.....

3- Find $f'(3)$ -using two points Centered finite-divided difference formula taking the data from the table.

Answer:

.....

End of questions

⇒ (Q.2) مثال : بسؤال مشت محدود توقف و بسبب الإمتحان محالين (تقريباً)
 ⇒ sol : المطلوب : كان ناسي (😊)
 ⇒ المطلوب : كان ناسي (😊)

$$\Rightarrow f(x) = x^2 - 4 \sin(5x) \quad / \quad [x_L = 1.3, x_u = 1.7]$$

(minimum) المطلوب

(Red) الأداة : المطلوب : كان ناسي (😊)

$$\Rightarrow d = 0.61803 (x_u - x_L) \Rightarrow d = 0.61803 (1.7 - 1.3)$$

$$\Rightarrow d = 0.247 \quad \times$$

$$\Rightarrow x_1 = x_L + d \Rightarrow x_1 = 1.3 + 0.247 \Rightarrow x_1 = 1.547 \quad \times$$

$$\Rightarrow x_2 = x_u - d \Rightarrow x_2 = 1.7 - 0.247 \Rightarrow x_2 = 1.453 \quad \times$$

$$\Rightarrow f(x_1) = (1.547)^2 - [4 \sin(5 \times 1.547)] \Rightarrow f(x_1) = -1.589 \quad \times$$

$$\Rightarrow f(x_2) = (1.453)^2 - [4 \sin(5 \times 1.453)] \Rightarrow f(x_2) = -1.215 \quad \times$$

(minimum) المطلوب : كان ناسي (😊)
 $\Rightarrow f(x_2) > f(x_1) \Rightarrow x_L = x_2 \text{ \& } x_u = x_u \quad \times$
 من الأداة

$$\Rightarrow \text{new interval} \rightarrow [x_L = 1.453, x_u = 1.7] \quad \times$$

هذا المثال
 2 iteration

$$\Rightarrow d = 0.61803 (x_u - x_L) \Rightarrow d = 0.61803 (1.7 - 1.453)$$

$$\Rightarrow d = 0.153 \quad \times$$

$$\Rightarrow x_1 = x_L + d \Rightarrow x_1 = 1.453 + 0.153 \Rightarrow x_1 = 1.606 \quad \times$$

$$\Rightarrow x_2 = x_u - d \Rightarrow x_2 = 1.7 - 0.153 \Rightarrow x_2 = 1.547 \quad \times$$

$$\Rightarrow f(x_1) = (1.606)^2 - [4 \sin(5 \times 1.606)] \Rightarrow f(x_1) = -1.359 \quad \times$$

$$\Rightarrow f(x_2) = (1.547)^2 - [4 \sin(5 \times 1.547)] \Rightarrow f(x_2) = -1.579 \quad \times$$

min ✗

$$\Rightarrow [f(x_1) > f(x_2)] \Rightarrow x_u = x_1 \text{ \& } x_l = x_2 \text{] ✗}$$

من اليسار

$$\Rightarrow \text{New Interval} \rightarrow [x_u = 1.606, x_l = 1.453] \text{ ✗}$$

$$\Rightarrow d = 0.61803 (x_u - x_l) \Rightarrow d = 0.61803 (1.606 - 1.453)$$

$$\Rightarrow \boxed{d = 0.095} \text{ ✗}$$

$$\Rightarrow x_1 = x_l + d \Rightarrow x_1 = 1.453 + 0.095 \Rightarrow \boxed{x_1 = 1.548} \text{ ✗}$$

$$\Rightarrow x_2 = x_u - d \Rightarrow x_2 = 1.606 - 0.095 \Rightarrow \boxed{x_2 = 1.511} \text{ ✗}$$

$$\Rightarrow f(x_1) = (1.548)^2 - [4 \sin(5 \times 1.548)] \Rightarrow \boxed{f(x_1) = -1.578} \text{ ✗}$$

$$\Rightarrow f(x_2) = (1.511)^2 - [4 \sin(5 \times 1.511)] \Rightarrow \boxed{f(x_2) = -1.539} \text{ ✗}$$

min

$$\Rightarrow [f(x_2) > f(x_1)] \Rightarrow x_l = x_2 \text{ \& } x_u = x_u \text{] ✗}$$

من اليسار

$$\Rightarrow \text{New Interval} \Rightarrow [x_l = 1.511, x_u = 1.606]$$

~~مع~~ ✗

~~بركيوك~~

Q.23. $\left\{ \int_0^1 \frac{3}{1+x^2} \cdot dx \right\}$

⇒ sol:-

$f(x) = \frac{3}{1+x^2}$ / $n=1, 2, 4, 8 \Rightarrow 3^{\text{rd}} \text{ order.}$

$\{1 - (n/1)\} \text{ error} \leftarrow \text{error}$

⇒ ~~$n=1$~~ $n=1$.

⇒ $h = \frac{b-a}{n} \Rightarrow h = \frac{1-0}{1} = \boxed{1}$ *

⇒ $R(1,0) = \frac{h}{2} [f(0) + f(1)]$

$= \frac{1}{2} [3 + 1.5]$

⇒ $R(1,0) = 2.25$ *

(i, k)	2 nd	2 nd	3 rd
$R(1,0)$	2.25	—	—
$R(2,0)$	$R(2,1)$	2.37	—
$R(3,0)$	$R(3,1)$	$R(3,2)$	—
$R(4,0)$	$R(4,1)$	$R(4,2)$	$R(4,3)$

لذلك نرى أن

(٢)

⇒ ~~$n=2$~~ $n=2$.

⇒ $h = \frac{b-a}{n} \Rightarrow h = \frac{1-0}{2} = \boxed{0.5}$ *

⇒ $R(2,0) = \frac{h}{2} [f(0) + f(1) + (2 \times f(0.5))]$

$= \frac{0.5}{2} [3 + 1.5 + (2 \times 2.4)]$

$R(2,0) = 2.325$ *

⇒ $n=4$.

⇒ $h = \frac{b-a}{n} \Rightarrow h = \frac{1-0}{4} = \boxed{0.25}$ *

⇒ $R(3,0) = \frac{h}{2} [f(0) + f(1) + 2(f(0.25) + f(0.5))]$

⇒ $R(3,0) = \frac{0.25}{2} [3 + 1.5 + 2(2.823 + 2.4)]$

⇒ $R(3,0) = \boxed{2.348}$ *

$$\Rightarrow \underline{n=8}$$

$$\Rightarrow h = \frac{b-a}{n} \Rightarrow h = \frac{1-0}{8} = \boxed{0.125} \times$$

$$\Rightarrow R(4,0) = \frac{h}{2} \left[f(0) + f(1) + 2 \left(f(0.125) + f(0.25) + f(0.375) + f(0.5) + f(0.625) + f(0.75) + f(0.875) \right) \right]$$

$$\Rightarrow R(4,0) = \frac{0.125}{2} \left[3 + 1.5 + 2(2.954 + 2.823 + 2.63 + 2.4 + 2.157 + 1.92 + 1.699) \right]$$

$$\Rightarrow R(4,0) = \boxed{2.354} \times$$

قانون روم - Rom

$$\Rightarrow R(2,1) = \frac{(4^k \times R(2,0)) - R(1,0)}{4^k - 1} \Rightarrow R(2,1) = \frac{(4 \times 2.325) - (2.35)}{3}$$

$$\Rightarrow \boxed{R(2,1) = 2.35} \times$$

$$\Rightarrow R(3,1) = \frac{(4^k \times R(3,0)) - R(2,0)}{4^k - 1} \Rightarrow R(3,1) = \frac{(4 \times 2.348) - (2.325)}{3}$$

$$\Rightarrow \boxed{R(3,1) = 2.356} \times$$

$$R(4,1) = \frac{(4^k \times R(4,0)) - R(3,0)}{4^k - 1} \Rightarrow R(4,1) = \frac{(4 \times 2.354) - 2.348}{3}$$

$$\Rightarrow \boxed{R(4,1) = 2.356} \times$$

$$\Rightarrow R(3,2) = \frac{(4^2 \times R(3,1)) - R(2,1)}{4^2 - 1}$$

$$\Rightarrow R(3,2) = \frac{(16 \times 2.356) - 2.35}{15} \Rightarrow \boxed{R(3,2) = 2.356} //$$

$$\Rightarrow R(4,2) = \frac{(4^2 \times R(4,1)) - R(3,1)}{15}$$

$$\Rightarrow R(4,2) = \frac{(16 \times 2.356) - (2.356)}{15} \Rightarrow \boxed{R(4,2) = 2.356} //$$

$$\Rightarrow R(4,3) = \frac{(4^3 \times R(4,2)) - R(3,2)}{63}$$

$$\Rightarrow R(4,3) = \frac{(64 \times 2.356) - 2.356}{63}$$

$$\Rightarrow R(4,3) = \boxed{2.356} //$$

Ibrahim Kh

Alc

[Q.3] .

⇒ 50/2

$$f(x_i, y_i)$$

$$\frac{2dy}{dx} - 3e^x + 2y = 0 \Rightarrow \left[\frac{dy}{dx} = \frac{3e^x - 2y}{2} \right]$$

$y(0) = 0 \xrightarrow{\text{logic}} \{x_i = 0, y_i = 0\} / h = 0.5 / y(1)$
 $y(0.5)$
 $y(1)$

$$\Rightarrow k_1 = h \cdot f(x_0, y_0) \Rightarrow k_1 = 0.5 \cdot f(0, 0) \Rightarrow k = 0.5 \cdot 1.5 \Rightarrow k_1 = 0.75$$

$$\Rightarrow k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) \Rightarrow k_2 = 0.5 f(0.25, 0.375) \Rightarrow k_2 = 0.5 * 1.551$$

$$\Rightarrow K_2 = 0.775 *$$

$$\Rightarrow k_3 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}) \Rightarrow k_3 = 0.5 f(0.25, 0.387) \Rightarrow k_3 = 0.5 * 4.539$$

$$\Rightarrow k_3 = 0.769, \#$$

$$\Rightarrow k_x = h f(x_0 + h, y_0 + k_3) \Rightarrow k_x = 0.5 f(0.5, 0.769) \Rightarrow k_x = 0.5 \cdot 1.704$$

$$\Rightarrow K_4 = 0.852 \quad \#$$

$$\Rightarrow y(0.5) = y(0) + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$\Rightarrow y(0.5) = 0 + \frac{1}{6} (0.75 + (2 \times 0.775) + (2 \times 0.769) + 0.852)$$

$y(0.5) = 0.782$ $\nRightarrow$ step (2) $\rightarrow$ done

Area

$$\Rightarrow \text{step (2)} \rightarrow [x_0 = 0.5 / y_0 = 0.782] \quad \#$$

$$\Rightarrow k_1 = h f(x_0, y_0) \Rightarrow k_1 = 0.5 f(0.5, 0.782) \Rightarrow k_1 = 0.5 \times 1.691$$

$$\Rightarrow \boxed{k_1 = 0.845} \quad \#$$

$$\Rightarrow k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) \Rightarrow k_2 = 0.5 f(0.75, 1.204)$$

$$\Rightarrow k_2 = 0.5 \times 1.971 \Rightarrow \boxed{k_2 = 0.985} \quad \#$$

$$\Rightarrow k_3 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) \Rightarrow k_3 = 0.5 f(0.75, 1.274)$$

$$\Rightarrow k_3 = 0.5 \times 1.901 \Rightarrow \boxed{k_3 = 0.950} \quad \#$$

$$\Rightarrow k_4 = h f(x_0 + h, y_0 + k_3) \Rightarrow k_4 = 0.5 f(1, 1.732)$$

$$\Rightarrow k_4 = 0.5 \times 2.345 \Rightarrow \boxed{k_4 = 1.172} \quad \#$$

$$\Rightarrow y(1) = y(0.5) + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$\Rightarrow y(1) = 0.782 + \frac{1}{6} (0.845 + (2 \times 0.985) + (2 \times 0.950) + 1.172)$$

$$\Rightarrow \boxed{y(1) = 1.730} \quad \# \Rightarrow \text{step (2)} \rightarrow \underline{\underline{\text{done}}}$$

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(Q.4)

x	-1	0	1	3	5
f(x)	3.1	0	4.3	7.5	2.2

(1). $\Rightarrow$ sol:-

x_i	y_i	b_1	b_2	b_3	b_4
-1	3.1	-3.1	3.7	-1.15	0.186
0	0	4.3	-0.9	-0.032	
1	4.3	1.6	-1.062		
3	7.5	-2.65			
5	2.2				

ملاحظات للتذكير:-
 III عند (3) نقال 3 ارقام اوقف
 عند الدرجة (4) $\leftarrow$ وبالتالي
 1 اوقف عند
(b4)

$$\Rightarrow b_{1(1)} = \frac{3.1 - 0}{-1 - 0} = -3.1$$

$$\Rightarrow b_{1(2)} = \frac{0 - 4.3}{0 - 1} = 4.3$$

$$\Rightarrow b_{1(3)} = \frac{4.3 - 7.5}{1 - 3} = 1.6$$

$$\Rightarrow b_{1(4)} = \frac{7.5 - 2.2}{3 - 5} = -2.65$$

$$\Rightarrow b_{2(1)} = \frac{-3.1 - 4.3}{-1 - 1} = 3.7$$

$$\Rightarrow b_{2(2)} = \frac{4.3 - 1.6}{0 - 3} = -0.9$$

$$\Rightarrow b_{2(3)} = \frac{1.6 - 2.65}{1 - 5} = -1.062$$

$$\Rightarrow b_{3(1)} = \frac{3.7 + 0.9}{-1 - 3} = -1.15$$

$$\Rightarrow b_{3(2)} = \frac{-0.9 + 1.062}{0 - 5} = -0.032$$

$$b_4 = \frac{-1.15 + 0.032}{-1 - 5}$$

$$b_4 = 0.186$$

$$\Rightarrow f(x) = b_0 + b_1(x-x_0) + b_2(x-x_0)(x-x_1) + b_3(x-x_0)(x-x_1)(x-x_2) \\ + b_4(x-x_0)(x-x_1)(x-x_2)(x-x_3)$$

دالة رانك

$$\Rightarrow f(2) = 3.1 + [-3.1(2+1)] + [3.7(2+1)(2-0)] \\ + [-1.15(2+1)(2-0)(2-1)] + [0.186(2+1)(2-0)(2-1)(2-3)]$$

$$\Rightarrow \boxed{f(2) = 7.984}$$

Ans

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⇒ (2).

⇒ solve

$$X = -1 \xrightarrow{to} X = 5$$

بقدره كذا زوجي
بقدره كذا زوجي

$$h = 5 - 1 = 4$$

عند (5) نقاء ← زرا ←

مشتق متساوية
زرا لازم كذا
النقاء
→ $\sin(\frac{1}{3})$ → حس → h →

$$\Rightarrow X = -1 \xrightarrow{to} X = 1 \rightarrow$$

عند 1 نقاء

$$[-1 \& 0 \& 1] \rightarrow$$

وملاحظة
(h) 1
هون متساوية
وتساوي = 1

$$\Rightarrow \left. \begin{array}{l} X_0 = -1 \\ X_1 = 0 \\ X_2 = 1 \end{array} \right\}$$

$$\Rightarrow I_1 = \frac{h}{3} [f(X_0) + f(X_2) + 4f(X_1)]$$

$$\Rightarrow I = \frac{1}{3} [3 \cdot 1 + 4 \cdot 3 + (4 \cdot 0)]$$

$$\Rightarrow I_1 = 2.467$$

الصور بيها
الجدول فمب زول
السؤال

حسا ← زحل نقطتان متوحد ← (Trap ---)

$$\Rightarrow X = 3 \& X = 5 \Rightarrow h = 2 \quad \begin{array}{l} X_0 = 3 \\ X_1 = 5 \end{array}$$

$$\Rightarrow I_2 = \frac{h}{2} [f(X_0) + f(X_1)]$$

$$= \frac{2}{2} [7.5 + 2.2]$$

$$= 9.7$$

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$$\Rightarrow I_{total} = I_1 + I_2 = 2.467 + 9.7$$

$$I_{total} = 12.167$$

(3). $\Rightarrow$ solve

$f'(3) \xrightarrow{\text{use}} \text{center \& first dir.}$ (مشتق أولي)

$$\Rightarrow f'(3) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h}$$

كيف بدى أعرفها $\leftarrow$ ما جينا هو الجدول لأننا هو الأساس (السؤال ومناه) ال (3) وقد غر بنحلوا.

$\Rightarrow$ هو بدى $\leftarrow f'(3) \leftarrow$ قيمة $\leftarrow$ ال (3) بدى عيين ومدين واقملا

1	3	5
4.3	7.5	2.2

$$\Rightarrow h = 2$$

$$\Rightarrow f'(3) = \frac{f(5) - f(1)}{2 \times 2} = \frac{2.2 - 4.3}{4}$$

$$\Rightarrow f'(3) = -0.525$$

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دعواتكم للأهلنا
في غزوة الحرة

